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Inverse substitutions

Composition with the identity substitution J changes no substitution. Thus, in the composition law, J is regarded as the unit element, and may be omitted whenever it occurs as a factor composed with other substitutions.

We now have the following theorem.

Theorem. For every substitution A there exists a unique inverse substitution A^{-1} satisfying

$$AA^{-1} = A^{-1}A = J. \tag{1}$$

This follows from the basic formulas of determinant theory. If the entries of A^{-1} are denoted by $\alpha_h^{(k)}$, then they are determined by the linear equations obtained from the requirement that the product with A be the identity. Since the determinant of A is assumed non-zero, these equations have a unique solution. The inverse substitution is therefore uniquely determined.

For inverses of composites one obtains, from

$$ABB^{-1}A^{-1} = J,$$

the formula

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{2}$$

If A, B, C are three substitutions, then from either of the equations

$$AB = AC, \qquad BA = CA,$$

composition with A^{-1} gives $B = C$. Thus the characteristic properties required for a group are satisfied under composition. The totality of all substitutions of a fixed dimension is therefore an infinite group. Moreover, if one selects from this totality any set closed under composition, then that set itself forms a group, finite or infinite according to the case.

Transformed substitutions

Let L be a fixed substitution. From any substitution A of the same dimension one obtains another substitution

$$A' = L^{-1}AL, \tag{3}$$

which is called the transform of A by L.

Putting

$$(x) = L(x'), \qquad (y) = L(y'), \tag{4}$$

one obtains, from $(y) = A(x)$,

$$(y') = A'(x'). \tag{5}$$

Consequently the passage to the transformed substitution is equivalent to applying the same change of variables, namely L^{-1}, to both systems of variables.

If

$$A' = L^{-1}AL, \qquad B' = L^{-1}BL,$$

then the laws of composition give

$$A'B' = L^{-1}ABL.$$

Hence:

Theorem. As A runs through the substitutions of a group, and L is kept fixed, the transformed substitutions $L^{-1}AL$ run through the substitutions of an isomorphic group.

Transposed substitutions

Inverse substitutions must be distinguished from transposed substitutions. From a substitution A one obtains a definite new substitution, denoted here by A_t, by turning rows into columns and columns into rows; equivalently, in Weber's notation, one interchanges the upper and lower indices of the coefficients of A.

If in the coefficient formula for the product $C = BA$ one interchanges upper and lower indices and simultaneously interchanges the letters a and b, one obtains the coefficients of the transpose of the product. Thus, when the factors in the product are transposed, their order is reversed:

$$(BA)_t = A_t B_t. \tag{6}$$

The operation of transposition is therefore formally similar to taking inverses, in that an order reversal occurs. Combining both operations gives

$$(A^{-1})_t = (A_t)^{-1}, \tag{7}$$

and, where no ambiguity is possible, this may be written simply as A_t^{-1}.

Linear fractional substitutions in one variable

We now apply the preceding general results to some special cases, beginning with substitutions of one variable. These substitutions are the same as the linear fractional functions

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \tag{8}$$

for which Weber introduces the abbreviation

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{9}$$

Their composition is governed by

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix} = \begin{pmatrix} \alpha\alpha' + \beta\gamma' & \alpha\beta' + \beta\delta' \\ \gamma\alpha' + \delta\gamma' & \gamma\beta' + \delta\delta' \end{pmatrix}. \tag{10}$$

The coefficients $\alpha, \beta, \gamma, \delta$ are arbitrary numbers subject only to the condition that

$$D = \alpha\delta - \beta\gamma \neq 0.$$

The identity substitution is

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{11}$$

and the inverse of (9) is

$$\begin{pmatrix} \dfrac{\delta}{D} & \dfrac{-\beta}{D} \\ \dfrac{-\gamma}{D} & \dfrac{\alpha}{D} \end{pmatrix}. \tag{12}$$

The substitution transposed to (9) is obtained by interchanging the diagonal entries and changing the signs of the off-diagonal entries:

$$\begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix}. \tag{13}$$

Equivalently, this is obtained by replacing

$$\alpha, \beta, \gamma, \delta \quad \text{by} \quad \delta, -\gamma, -\beta, \alpha.$$

One verifies immediately that

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} = \begin{pmatrix} D & 0 \\ 0 & D \end{pmatrix}. \tag{14}$$

This both confirms the preceding rule for transposes and shows that every such substitution commutes with its transposed substitution.

Permutation groups as linear substitutions

Let $x_1, x_2, \ldots, x_n$ be a system of n variables, and let

$$\alpha_1, \alpha_2, \ldots, \alpha_n$$

be some ordering of the numbers $1, 2, \ldots, n$. The equations

$$x_1 = x'_{\alpha_1}, \quad x_2 = x'_{\alpha_2}, \quad \ldots, \quad x_n = x'_{\alpha_n} \tag{15}$$

define a linear substitution of dimension n,

$$A = \begin{pmatrix} a_1^{(1)} & \cdots & a_n^{(1)} \\ \cdots & \cdots & \cdots \\ a_1^{(n)} & \cdots & a_n^{(n)} \end{pmatrix}, \qquad (x) = A(x'), \tag{16}$$

in which every row and every column contains exactly one non-zero coefficient, and that coefficient is 1. Hence $|A| = \pm 1$. If, after carrying out the substitution, one writes x_i again in place of x'_i, the result is just the permutation

$$\begin{pmatrix} 1, 2, \ldots, n \\ \alpha_1, \alpha_2, \ldots, \alpha_n \end{pmatrix}$$

of the indices.

Let B be another such substitution,

$$(x') = B(x''), \tag{17}$$

or, more explicitly,

$$x'_1 = x''_{\beta_1}, \quad x'_2 = x''_{\beta_2}, \quad \ldots, \quad x'_n = x''_{\beta_n}. \tag{18}$$

Then composition gives

$$x_1 = x''_{\beta_{\alpha_1}}, \quad x_2 = x''_{\beta_{\alpha_2}}, \quad \ldots, \quad x_n = x''_{\beta_{\alpha_n}}, \tag{19}$$

which is abbreviated by

$$(x) = AB(x''). \tag{20}$$

In other words,

$$\begin{pmatrix} 1, 2, \ldots, n \\ \alpha_1, \alpha_2, \ldots, \alpha_n \end{pmatrix} \begin{pmatrix} 1, 2, \ldots, n \\ \beta_1, \beta_2, \ldots, \beta_n \end{pmatrix} = \begin{pmatrix} 1, 2, \ldots, n \\ \beta_{\alpha_1}, \beta_{\alpha_2}, \ldots, \beta_{\alpha_n} \end{pmatrix}. \tag{21}$$

Thus, when the substitutions A and B are regarded as permutations of the indices, their product as substitutions agrees with their composite as permutations.

Therefore permutation groups on n letters are simply a special case of finite groups of linear substitutions of dimension n. Even permutations correspond to substitutions of determinant $+1$, and odd permutations to substitutions of determinant -1.

This connection between permutations and linear substitutions makes it possible to derive from every permutation group a group of linear substitutions, and conversely to treat certain questions about permutation groups by means of the theory of linear substitutions.

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1 The extended notion of invariant

Relative invariants are in essence a special case of a more general notion, which, in a manner similar to that of relative invariants, can be used for reducing the form problem.

We look for systems of homogeneous forms of the same degree in the variables

$$x_1, x_2, \ldots, x_n,$$

namely

$$X_1, X_2, \ldots, X_m, \tag{22}$$

with the property that, when the substitutions of the group S are applied, the functions $X_1, \ldots, X_m$ undergo a system Σ of linear substitutions. These substitutions Σ then themselves form a group; this group is isomorphic to S, either simply or with several stages of identification. If $m = 1$, one recovers the notion of a relative invariant.

We now seek all substitutions of S which leave each of the functions (22) individually unchanged. These substitutions form a group, which we denote by T, and we shall show that T is a normal divisor of S. Let s be a substitution in S which induces the substitution σ on the X's. Then s^{-1} induces σ^{-1}. If τ is a substitution in T, then τ leaves all the X's fixed. Therefore $s^{-1}\tau s$ induces

$$\sigma^{-1}\sigma = 1$$

on the X's, and hence also leaves all the X's fixed. Thus $s^{-1}\tau s$ belongs to T, and T is a normal divisor of S.

Let μ and ν be the orders of S and T, and write

$$\mu = e\nu.$$

Then e is the index of T and at the same time the order of the group Σ.

Every absolute invariant of the group T, that is, every function of x which admits the substitutions of T, can be expressed rationally over Ω in terms of the X's.

The proof is the same argument that has already been used several times. Let ϑ be a function of the X's which assumes the e distinct values

$$\vartheta, \vartheta_1, \dots, \vartheta_{e-1},$$

and put

$$\Phi(t) = (t-\vartheta)(t-\vartheta_1)\cdots(t-\vartheta_{e-1}). \tag{23}$$

Then $\Phi(t)$ is a rational function of the variable t over Ω.

Let r be an absolute invariant of T. It assumes at most e different values corresponding to $\vartheta, \vartheta_1, \dots, \vartheta_{e-1}$, say

$$r, r_1, \dots, r_{e-1},$$

where repetitions may occur. Then the function

$$\frac{\Phi(t)r}{t-\vartheta} + \frac{\Phi(t)r_1}{t-\vartheta_1} + \cdots + \frac{\Phi(t)r_{e-1}}{t-\vartheta_{e-1}} = F(t)$$

belongs to Ω, and for $t = \vartheta$ one obtains

$$r = \frac{F(\vartheta)}{\Phi'(\vartheta)}.$$

Since $\Phi'(\vartheta) \neq 0$, this expresses r rationally in ϑ. Hence r belongs to the field

$$\Omega(X_1, X_2, \dots, X_m).$$

The invariants of T are also invariants of S. After the form problem for the group Σ has been solved, the functions $X_1, \dots, X_m$ are known, and therefore the invariants of T are known as well. Thus the form problem for S is reduced to the successive solution of the two form problems belonging to the groups Σ and T, both of lower degree than the original form problem for S. A form problem which can be decomposed in this way into two lower-degree form problems will be called *imprimitive.* Since such a reduction requires the existence of a normal divisor T of S distinct from both S and the identity, it follows that, if S is simple, the corresponding form problem is necessarily *primitive.*

Conversely, if S has a normal divisor of index e, then the form problem is imprimitive. To obtain a system of functions $X_1, \dots, X_m$, one takes an invariant ϑ of T which assumes e different values under S, and one may take these values

$$\vartheta, \vartheta_1, \dots, \vartheta_{e-1}$$

as the functions $X_1, \dots, X_m$. The group Σ is then the permutation group induced by S on these ϑ's. In the sense of the preceding section, however, one wants m to be as small as possible, and a useful reduction of the problem is achieved only when $m < n$.

2 Normal forms

Before turning to special applications, we need one more general observation. We have already seen that from any group S of linear substitutions one obtains infinitely many isomorphic groups by transforming with an arbitrary substitution L of the same dimension, namely by forming

$$LSL^{-1}. \tag{24}$$

This is equivalent to replacing the variables x by variables y through

$$(x) = L(y). \tag{25}$$

This transformation can be used to put the group S into a simpler normal form; then the invariants also acquire definite normal forms, which in many cases may be quite simple.

For the normal form, let the resolvent of the form problem be written

$$\Phi(\vartheta, A_1, A_2, \ldots) = 0, \tag{26}$$

where $A_1, A_2, \ldots$ denote invariants of the group S. The variables x_i can then be expressed rationally in terms of ϑ and the A_i:

$$x_i = \varphi_i(\vartheta, A_1, A_2, \ldots). \tag{27}$$

Once the function ϑ has been fixed, these expressions take a definite form for the normal form of the group.

It may also happen that the invariants are not given in the normal form, but in some other form, which we shall call the *general form.* One then has to determine the substitution L which transforms the general form into a previously recognized possible normal form. This task is no other than the form problem itself for the normal form.

Assume that the invariants in the general form, considered as functions of y, are $B_1, B_2, \ldots$, so that the substitution (25) produces the identities

$$A_1 = B_1, \qquad A_2 = B_2, \qquad \ldots. \tag{28}$$

The task is to determine the substitution L making these equations identities, when the forms of the functions A_i and B_i are known. This is solved once one assumes solved equation (26) for a suitably chosen special system of values of the A_i.

Indeed, differentiate (26) and (27):

$$\begin{aligned} 0 &= \Phi'(\vartheta)d\vartheta + \sum_i \Phi'(A_i)dA_i, \\ dx_i &= \varphi_i'(\vartheta)d\vartheta + \sum_i \varphi_i'(A_i)dA_i. \end{aligned}$$

Eliminating $d\vartheta$ gives

$$dx_i = \sum_i \frac{\varphi_i'(A_i)\Phi'(\vartheta) - \Phi'(A_i)\varphi_i'(\vartheta)}{\Phi'(\vartheta)}\, dA_i. \tag{29}$$

In consequence of (28), one has

$$dA_i = \sum_k \frac{\partial B_i}{\partial y_k}\, dy_k. \tag{30}$$

Comparison with the differentials of the substitution $x = L(y)$ then determines the coefficients of L as rational expressions once a special value system for the y's has been chosen, subject only to the condition that $\Phi'(\vartheta)$ not vanish. Thus the determination of the normalizing substitution has been reduced to the same resolvent which solves the form problem.

3 Ternary orthogonal substitutions

We now have to pick out, from the totality of linear substitutions, narrower groups, and finally finite groups. Such narrower groups are obtained by imposing the condition that certain homogeneous functions of the variables should remain invariant. Instead of treating this in full generality, we pass immediately to the most important special case. We restrict ourselves to ternary substitutions and require that a quadratic form of non-vanishing determinant remain invariant. Since every such quadratic form can be transformed linearly into a sum of squares, no generality is lost by taking the form to be the sum of squares. Such substitutions are called *orthogonal.*

Thus

$$(y_1, y_2, y_3) = A(x_1, x_2, x_3) \tag{31}$$

is orthogonal if its coefficients are chosen so that the identity

$$y_1^2 + y_2^2 + y_3^2 = x_1^2 + x_2^2 + x_3^2 \tag{32}$$

holds. Substituting the expressions (31) into (32) and equating corresponding coefficients yields six relations among the nine coefficients of A.

If

$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}, \tag{33}$$

these relations are

$$\begin{aligned} a_1^2 + b_1^2 + c_1^2 = 1, &\quad a_2a_3 + b_2b_3 + c_2c_3 = 0, \\ a_2^2 + b_2^2 + c_2^2 = 1, &\quad a_3a_1 + b_3b_1 + c_3c_1 = 0, \\ a_3^2 + b_3^2 + c_3^2 = 1, &\quad a_1a_2 + b_1b_2 + c_1c_2 = 0. \end{aligned} \tag{34}$$

They are the familiar relations of analytic geometry. From them, using the multiplication theorem for determinants, one obtains $|A|^2 = 1$, and therefore $|A| = \pm 1$.

Equations (34) also show that the inverse substitution of A is identical with the transpose:

$$A^{-1} = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}.$$

This property could itself serve as the definition of orthogonal substitutions.

All orthogonal substitutions form a group. Inside it lies the narrower group distinguished by

$$|A| = +1, \tag{35}$$

which Weber calls the group of *proper orthogonal substitutions.*

The substitution (31) may be interpreted as the passage from one rectangular coordinate system to another with the same origin. If the determinant condition (35) holds, then the first coordinate system can be brought into coincidence with the second by a rotation about a fixed axis through a definite angle.

To see this, set

$$D = \begin{vmatrix} a_1 - 1 & a_2 & a_3 \\ b_1 & b_2 - 1 & b_3 \\ c_1 & c_2 & c_3 - 1 \end{vmatrix}.$$

From the orthogonality relations one obtains $|A|D = -D$; hence, when $|A| = +1$, $D = 0$. Therefore one can determine λ, μ, ν from

$$\begin{aligned} \lambda &= a_1\lambda + a_2\mu + a_3\nu, \\ \mu &= b_1\lambda + b_2\mu + b_3\nu, \\ \nu &= c_1\lambda + c_2\mu + c_3\nu, \\ \lambda^2 + \mu^2 + \nu^2 &= 1. \end{aligned} \tag{36}$$

These numbers determine the direction of a straight line making with the axes y_1, y_2, y_3 the same angles as with x_1, x_2, x_3; this line is the axis of rotation. In the case $|A| = -1$ this conclusion no longer applies.

There is also another interpretation: (x) and (y) may be regarded as the coordinates of two different points, referred to the same coordinate system. If x describes a line, surface, or body, then y describes a congruent figure; when $|A| = -1$, it is the mirror image.

The group of proper orthogonal substitutions is equivalent to the group obtained from the possible positions of a rigid body rotating about a fixed point. If a position E is chosen as the identity, every other position A is reached by a rotation about a certain axis through a certain angle. If B is another position, then the compound position AB is the position obtained by carrying out the rotation which leads to A, but starting from the position B rather than from the identity. This is exactly the composition law for the corresponding orthogonal substitutions.

By composing the proper orthogonal substitutions with one improper substitution, for example the reflection

$$(x_1, x_2, x_3) = (y_1, y_2, -y_3),$$

one obtains all orthogonal substitutions.

The finite groups contained in the group of proper orthogonal substitutions are of special interest. Geometrically, such a finite group of degree g corresponds to a finite set of positions of a body. If one imagines the body fixed simultaneously in all these positions and unites the whole into a new rigid body, one obtains a figure that can cover itself in g different positions. The clearest cases occur when the body is bounded by planes: regular pyramids, double pyramids, and the regular solids.

A regular pyramid with g sides covers itself in g positions by rotations about its principal axis. A regular double pyramid has additional half-turn symmetries about axes in the equatorial plane. The tetrahedron has 12 such positions, the cube and octahedron have 24, and the dodecahedron and icosahedron have 60. In each case the number is equivalently the product of the number of vertices with the number of edges or faces meeting at a vertex, or the product of the number of faces with the number of edges or vertices of one face, or twice the number of edges.

4 Linear fractional substitutions

The group of ternary orthogonal substitutions is isomorphic to the group of linear fractional substitutions of one variable, or equivalently to the binary substitutions of ratios.

Let

$$(y_1, y_2, y_3) = A(x_1, x_2, x_3), \tag{37}$$

$$y_1^2 + y_2^2 + y_3^2 = x_1^2 + x_2^2 + x_3^2. \tag{38}$$

Apply the fixed, non-orthogonal substitution

$$L = \begin{pmatrix} i & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & i/\sqrt{2} & -i/\sqrt{2} \end{pmatrix}, \quad L^{-1} = \begin{pmatrix} -i & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix}, \tag{39}$$

where $i = \sqrt{-1}$. Set

$$(y_1, y_2, y_3) = L(y_1', y_2', y_3'), \qquad (x_1, x_2, x_3) = L(x_1', x_2', x_3'), \tag{40}$$

$$(y_1', y_2', y_3') = A'(x_1', x_2', x_3'), \tag{41}$$

where

$$A' = L^{-1} A L. \tag{42}$$

Then (38) becomes

$$-y_1'^2 + 2y_2'y_3' = -x_1'^2 + 2x_2'x_3'. \tag{43}$$

Put

$$x_1' = \sqrt{2}\xi_1\xi_2, \qquad x_2' = \xi_1^2, \qquad x_3' = \xi_2^2. \tag{44}$$

Then $x_1'^2 - 2x_2'x_3'$ vanishes identically, and the images y_1', y_2', y_3' become binary quadratic forms in ξ_1, ξ_2 satisfying

$$y_1'^2 - 2y_2'y_3' = 0. \tag{45}$$

It follows that $y_2'y_3'$ is a perfect square. The forms y_2' and y_3' cannot have a common factor; otherwise the determinant of A' would vanish. Hence y_2' and y_3' must themselves be squares of linear forms. Write

$$\eta_1 = \alpha\xi_1 + \beta\xi_2, \qquad \eta_2 = \gamma\xi_1 + \delta\xi_2. \tag{46}$$

Then one may put

$$y_1' = \sqrt{2}\eta_1\eta_2, \qquad y_2' = \eta_1^2, \qquad y_3' = \eta_2^2. \tag{47}$$

Expanding and reverting from the ξ's to x_1', x_2', x_3', one obtains

$$A' = \begin{pmatrix} \alpha\delta + \beta\gamma & \sqrt{2}\alpha\gamma & \sqrt{2}\beta\delta \\ \sqrt{2}\alpha\beta & \alpha^2 & \beta^2 \\ \sqrt{2}\gamma\delta & \gamma^2 & \delta^2 \end{pmatrix}. \tag{48}$$

The condition (43) requires

$$\alpha\delta - \beta\gamma = \pm 1. \tag{49}$$

The plus sign corresponds to proper orthogonal substitutions and the minus sign to improper ones.

Thus A' is determined by the binary substitution

$$(\eta_1, \eta_2) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} (\xi_1, \xi_2), \qquad \alpha\delta - \beta\gamma = \pm 1, \tag{50}$$

or, in terms of ratios $\eta = \eta_1/\eta_2$, $\xi = \xi_1/\xi_2$, by the linear fractional substitution

$$\eta = \frac{\alpha\xi + \beta}{\gamma\xi + \delta}. \tag{51}$$

The four coefficients $\alpha, \beta, \gamma, \delta$ determine the same ternary substitution after simultaneous multiplication by -1. Thus the ternary substitution corresponds not to a single binary matrix but to a pair of opposite binary matrices, or equivalently to the induced fractional-linear transformation.

Moreover, if A'', B'' are two such binary substitutions and A', B' the associated ternary substitutions, then the ternary substitution associated with $A''B''$ is $A'B'$. Hence the corresponding groups are isomorphic.

5 Reality conditions

We must still answer one question. Up to now no attention has been paid to the reality of the coefficients. If a geometric application is intended, the orthogonal substitution A must be real. We must therefore determine the conditions on $\alpha, \beta, \gamma, \delta$ which make the coefficients of A real.

Using the expression $A = LA'L^{-1}$ one obtains

$$A = \begin{pmatrix} \alpha\delta + \beta\gamma & i(\alpha\gamma + \beta\delta) & \alpha\gamma - \beta\delta \\ -i(\alpha\beta + \gamma\delta) & \dfrac{\alpha^2 + \beta^2 + \gamma^2 + \delta^2}{2} & \dfrac{-\alpha^2 + \beta^2 - \gamma^2 + \delta^2}{2} \\ \alpha\beta - \gamma\delta & i\dfrac{\alpha^2 - \beta^2 - \gamma^2 + \delta^2}{2} & \dfrac{\alpha^2 - \beta^2 + \gamma^2 - \delta^2}{2} \end{pmatrix}.$$

The entries of this matrix are to be real and

$$\alpha\delta - \beta\gamma = \pm 1. \tag{52}$$

The result is the following necessary and sufficient condition. The orthogonal substitution is real precisely when α and $\pm\delta$, and β and $\mp\gamma$, form two conjugate-imaginary pairs; the upper signs hold for proper orthogonal substitutions, and the lower signs for improper ones.

6 Finite groups of linear fractional substitutions and their poles

If all finite groups of linear fractional substitutions are known, then all finite groups of proper ternary orthogonal substitutions and all finite groups of binary linear substitutions of determinant 1 are thereby known as well.

Let G be a finite group of degree n whose elements are

$$\Theta_0(x) = x, \quad \Theta_1(x), \quad \dots, \quad \Theta_{n-1}(x), \tag{53}$$

where each Θ denotes a linear fractional function

$$\Theta(x) = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{54}$$

By transforming with an arbitrary linear substitution L, one obtains an isomorphic group

$$1, \quad L\Theta_1 L^{-1}, \quad \dots, \quad L\Theta_{n-1}L^{-1}.$$

The classification is considered complete once one representative from each such transformed family is found.

Since G is finite, every element Θ has a finite order: there is a least positive integer t such that $\Theta^t = 1$, and t divides n. For every non-identity substitution we consider the values of x fixed by the substitution:

$$x = \Theta(x). \tag{55}$$

These fixed values are called the *poles* of Θ. Inserting (54), one obtains the quadratic equation

$$\gamma x^2 + (\delta - \alpha)x - \beta = 0. \tag{56}$$

The roots coincide only in the exceptional case

$$(\delta - \alpha)^2 + 4\beta\gamma = 0, \tag{57}$$

or equivalently

$$(\alpha + \delta)^2 = 4\Delta, \qquad \Delta = \alpha\delta - \beta\gamma. \tag{58}$$

7 The possible types of groups

Let a be a pole of the group G, and suppose that there are exactly $\nu - 1$ non-identity elements

$$\Theta_1, \Theta_2, \dots, \Theta_{\nu-1}$$

of G such that

$$a = \Theta_1(a) = \Theta_2(a) = \cdots = \Theta_{\nu-1}(a). \tag{59}$$

Such a pole is called a ν-fold pole. The elements

$$1, \Theta_1, \Theta_2, \dots, \Theta_{\nu-1} \tag{60}$$

form a subgroup Q of G, because if both Θ_1 and Θ_2 fix a, then so does their product. Hence ν divides n:

$$n = \nu\mu. \tag{61}$$

This group Q is cyclic. After transforming a to ∞, the substitutions fixing it take the form

$$x, \quad \varepsilon_1 x + c_1, \quad \varepsilon_2 x + c_2, \quad \dots,$$

and the composition laws show that the ε_i are the ν-th roots of unity. No two of them can be equal without making the corresponding substitutions identical. Hence the group is generated by an element whose multiplier is a primitive ν-th root of unity:

$$Q = \{1, \Theta, \Theta^2, \dots, \Theta^{\nu-1}\}. \tag{62}$$

Thus:

Theorem. *A ν-fold pole determines a cyclic subgroup Q of degree ν contained in G.*

Theorem. *The two poles of any substitution Θ of the group G are poles of the same multiplicity, and they are the only poles of the corresponding cyclic group Q.*

Choose representatives $\Psi_1, \dots, \Psi_{\mu-1}$ for the non-trivial right cosets of Q, so that

$$G = Q + \Psi_1 Q + \cdots + \Psi_{\mu-1} Q. \tag{63}$$

Put

$$a_1 = \Psi_1(a), \quad \dots, \quad a_{\mu-1} = \Psi_{\mu-1}(a). \tag{64}$$

Then

$$a, a_1, \dots, a_{\mu-1}$$

are distinct and are all ν-fold poles. They form a system of conjugate ν-fold poles of G.

Every substitution χ of G permutes these poles. Hence two systems of conjugate poles are either identical or disjoint. Therefore all poles of G can be arranged into systems of conjugate poles.

Counting every ν-fold pole with multiplicity $\nu - 1$, the total number of poles is $2n - 2$. Thus

$$2n - 2 = \mu(\nu - 1) + \mu'(\nu' - 1) + \mu''(\nu'' - 1) + \cdots . \tag{65}$$

If h is the number of systems of conjugate poles, and $n = \mu\nu = \mu'\nu' = \cdots$, this becomes

$$2n - 2 = nh - \mu - \mu' - \mu'' - \cdots . \tag{66}$$

Since every ν is at least 2, one obtains

$$2 \le h \le 4 - \frac{4}{n}.$$

Therefore h is only 2 or 3.

If $h = 2$, then $\mu = \mu' = 1$ and $\nu = \nu' = n$, giving the cyclic case. If $h = 3$, one obtains exactly the following possibilities:

I. **Cyclic group.**

$$\nu = \nu' = n, \qquad \mu = \mu' = 1.$$

II. **Dihedral group.**

$$\nu = \nu' = 2, \qquad \nu'' = m, \qquad n = 2m, \qquad \mu = \mu' = m, \qquad \mu'' = 2.$$

III. **Tetrahedral group.**

$$\nu = 2, \qquad \nu' = 3, \qquad \nu'' = 3, \qquad n = 12.$$

IV. **Octahedral group.**

$$\nu = 2, \qquad \nu' = 3, \qquad \nu'' = 4, \qquad n = 24.$$

V. **Icosahedral group.**

$$\nu = 2, \qquad \nu' = 3, \qquad \nu'' = 5, \qquad n = 60.$$

These exhaust the possibilities. It remains only to show that groups of all five types actually exist.

8 Basic forms

To every system of conjugate poles of the group G there corresponds an invariant binary form. If the poles are

$$a, a_1, \ldots, a_{\mu-1},$$

then the corresponding form has roots at precisely these poles; in homogeneous variables it is

$$f(x_1, x_2) = \prod_j (x_1 - a_j x_2), \tag{67}$$

up to a constant factor. Under a substitution of the group the roots are merely permuted, so f is transformed into itself up to a constant factor.

Thus:

Theorem. *Every system of conjugate poles of G gives an invariant form whose degree is the number of poles in the system and whose roots are exactly those poles.*

These invariant forms will be called the *basic forms* of the group.

If $F(x_1, x_2)$ is any invariant form of G and shares a root with a basic form f, then F shares all the roots of f; hence F is divisible by f.

Consequently, if an invariant form F has degree less than the order of G, it must be, up to a constant factor, a product of powers of the basic forms. Equivalently: an invariant form of degree smaller than the order of the group which is not such a product must vanish identically.

9 Cyclic and dihedral groups

We now apply the preceding principles to construct the polyhedral groups. For a cyclic group of degree n, there are only two systems of conjugate poles, each consisting of a single $(n-1)$-fold pole. These may be taken to be 0 and ∞, giving the linear basic forms

$$f_1 = x_1, \qquad f_2 = x_2. \tag{68}$$

All substitutions are multiplicative, and their multipliers are n-th roots of unity. Thus, for a primitive n-th root ε, the group is

$$C_n = \left\{ \begin{pmatrix} \varepsilon^r & 0 \\ 0 & 1 \end{pmatrix} : r = 0, 1, \ldots, n-1 \right\}, \tag{69}$$

or, with determinant 1,

$$C_n = \left\{ \begin{pmatrix} \varepsilon^{r/2} & 0 \\ 0 & \varepsilon^{-r/2} \end{pmatrix} : r = 0, 1, \ldots, n-1 \right\}. \tag{70}$$

For the dihedral group D_m of degree $2m$, there are two systems of m conjugate two-fold poles and one system of two conjugate m-fold poles. Taking the latter to be 0 and ∞, the quadratic basic form is

$$f_1 = x_1 x_2.$$

The group consists of

$$\begin{aligned} &x, \varepsilon x, \varepsilon^2 x, \ldots, \varepsilon^{m-1} x, \\ &\frac{1}{x}, \frac{\varepsilon}{x}, \frac{\varepsilon^2}{x}, \ldots, \frac{\varepsilon^{m-1}}{x}, \end{aligned}$$

where ε is a primitive m-th root of unity. In matrix form,

$$D_m = \begin{pmatrix} \varepsilon^r & 0 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & \varepsilon^r \\ 1 & 0 \end{pmatrix}, \qquad r = 0, 1, \ldots, m-1. \tag{71}$$

With determinant 1, this may be written

$$\begin{pmatrix} \varepsilon^{r/2} & 0 \\ 0 & \varepsilon^{-r/2} \end{pmatrix}, \qquad \begin{pmatrix} 0 & i\varepsilon^{r/2} \\ i\varepsilon^{-r/2} & 0 \end{pmatrix}. \tag{72}$$

The remaining poles are obtained from

$$x = \frac{\varepsilon^h}{x},$$

and give the two basic forms

$$f_2 = x_1^m + x_2^m, \qquad f_3 = x_1^m - x_2^m. \tag{73}$$

10 The tetrahedral group

For the tetrahedral group there is a system of six conjugate two-fold poles. We may assume that the group contains

$$\Theta(x) = -x, \qquad \psi(x) = \frac{1}{x},$$

and hence also

$$\psi_1(x) = \psi\Theta(x) = -\frac{1}{x}.$$

The poles of these substitutions are $0, \infty, \pm 1, \pm i$. Therefore the equation of the two-fold poles is $x(x^4 - 1) = 0$, and the first basic form of degree 6 is

$$f(x_1, x_2) = x_1 x_2 (x_1^4 - x_2^4). \tag{74}$$

There must also be a substitution χ carrying -1 to 0 and $+1$ to ∞, and hence it has the form

$$\chi = \lambda \frac{x+1}{x-1}. \tag{75}$$

Requiring it to permute the six poles forces $\lambda = \pm i$. Choosing the upper sign, one obtains

$$\chi(x) = i\frac{x+1}{x-1}.$$

Then

$$\chi^2(x) = \frac{x+i}{x-i}, \qquad \chi^3(x) = x,$$

and the twelve substitutions of the tetrahedral group are

$$\begin{array}{cccc} 1, & \Theta, & \psi, & \Theta\psi, \\ \chi, & \Theta\chi, & \psi\chi, & \Theta\psi\chi, \\ \chi^2, & \Theta\chi^2, & \psi\chi^2, & \Theta\psi\chi^2. \end{array} \tag{76}$$

Equivalently they are

$$\pm x, \quad \pm\frac{1}{x}, \quad \pm i\frac{x+1}{x-1}, \quad \pm i\frac{x-1}{x+1}, \quad \pm\frac{x+i}{x-i}, \quad \pm\frac{x-i}{x+i}. \tag{77}$$

They satisfy the defining relations listed in the source, and therefore form the tetrahedral group generated by Θ, ψ, χ.

After a change of variables, the remaining basic forms can be written

$$\begin{aligned} \Phi_1 &= x_1^4 + 2\sqrt{-3}\, x_1^2 x_2^2 + x_2^4, \\ \Phi_2 &= x_1^4 - 2\sqrt{-3}\, x_1^2 x_2^2 + x_2^4. \end{aligned} \tag{78}$$

Eliminating x_1, x_2 from these formulas gives

$$12\sqrt{-3}\, f^2 = \Phi_1^3 - \Phi_2^3. \tag{79}$$

The determinant-1 representatives of Θ, ψ, χ are

$$\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad \begin{pmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & \frac{1+i}{2} \end{pmatrix}. \tag{80}$$

They satisfy the reality conditions above, so the corresponding ternary orthogonal group is real.

The Hessian of the basic form f is again an invariant:

$$\begin{aligned} H &= -\frac{1}{25}\left(f''(x_1, x_1) f''(x_2, x_2) - f''(x_1, x_2)^2\right) \\ &= (x_1^4 + x_2^4)^2 + 12 x_1^4 x_2^4, \end{aligned}$$

and is the product of the two functions Φ_1 and Φ_2.

11 The octahedral group

For the octahedral group, the general classification gives systems of conjugate poles of multiplicities $2, 3, 4$ and group order 24. Weber constructs it by adjoining to the tetrahedral group a further substitution that enlarges the order from 12 to 24.

Let the tetrahedral group be generated by substitutions Θ, ψ, χ satisfying

$$\begin{aligned} \Theta^2 = \psi^2 = 1, &\qquad \chi^3 = 1, \\ \chi\Theta = \Theta\psi\chi, &\qquad \chi\psi = \Theta\chi, \\ \chi^2\Theta = \psi\chi^2, &\qquad \chi^2\psi = \Theta\psi\chi^2. \end{aligned} \tag{81}$$

One obtains a group of order 24 with elements of the form

$$\chi^\lambda \omega^\mu \Theta^\nu,$$

where χ has order 3, ω has order 2, and Θ has order 4, subject to the relations displayed in Weber's construction. The resulting group is isomorphic to the octahedral group.

The remaining basic forms of degrees 8 and 12 are obtained from the tetrahedral basic form by covariant operations. One obtains

$$W = x_1^8 + 14 x_1^4 x_2^4 + x_2^8, \tag{82}$$

and from the functional determinant of f and W,

$$K = x_1^{12} - 33 x_1^8 x_2^4 - 33 x_1^4 x_2^8 + x_2^{12}. \tag{83}$$

12 The icosahedral group

For the icosahedral group there is a system of conjugate five-fold poles. We may therefore assume that the group contains

$$\Theta(x) = \varepsilon x, \qquad \psi(x) = -\frac{1}{x}, \tag{84}$$

where ε is a primitive fifth root of unity. The basic form of degree 12 associated with the five-fold poles must then have the form

$$f(x_1, x_2) = x_1 x_2 (x_1^{10} + m x_1^5 x_2^5 - x_2^{10}), \tag{85}$$

and the problem is to determine the constant m. The two other basic forms have degrees 20 and 30.

The constant m can be determined by forming a covariant of degree lower than the order 60 of the group which cannot be expressed as a product of powers of the three basic forms of degrees $12, 20, 30$; by the preceding divisibility theorem, such a covariant must vanish identically. Weber describes this through polar and Hessian operations on the binary form f.